

Assignment 2: Git and Stream Temperature

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Project Overview

This project analyzed long-term stream temperature trends at three monitoring sites on the Willamette River using USGS daily temperature data collected from 2010-2023.

Trend analyses were conducted separately for July and August temperatures using generalized liner models (GLMs). The goal was to determine whether stream temperatures are changing over time and whether warming trends differ across sites.

Data Loading

```
library(tidyverse)
library(lubridate)

# Load data
temp_data <- read.csv("data/temp_data.csv")

# Format variables
temp_data <- temp_data %>%
  select(
    agency_cd = monitoring_location_id,
    site_no = monitoring_location_id,
    Date = time,
    temp_C = value )

temp_data$site_no <- gsub("USGS-", "", temp_data$site_no)

temp_data$Date <- as.Date(temp_data$Date)
```

```
temp_data$month <- month(temp_data$Date, label = TRUE)
temp_data$year <- year(temp_data$Date)
```

Key Findings

Both July and August temperatures showed significant warming trends over time.

- July temperatures increased by approximately 0.15 °C per year at the baseline site.
- August temperatures increased by approximately 0.13 °C per year at the baseline site.
- Site 14211010 consistently showed stronger warming trends than the other sites.
- Site 14211720 did not show significantly different warming trends from the baseline site.

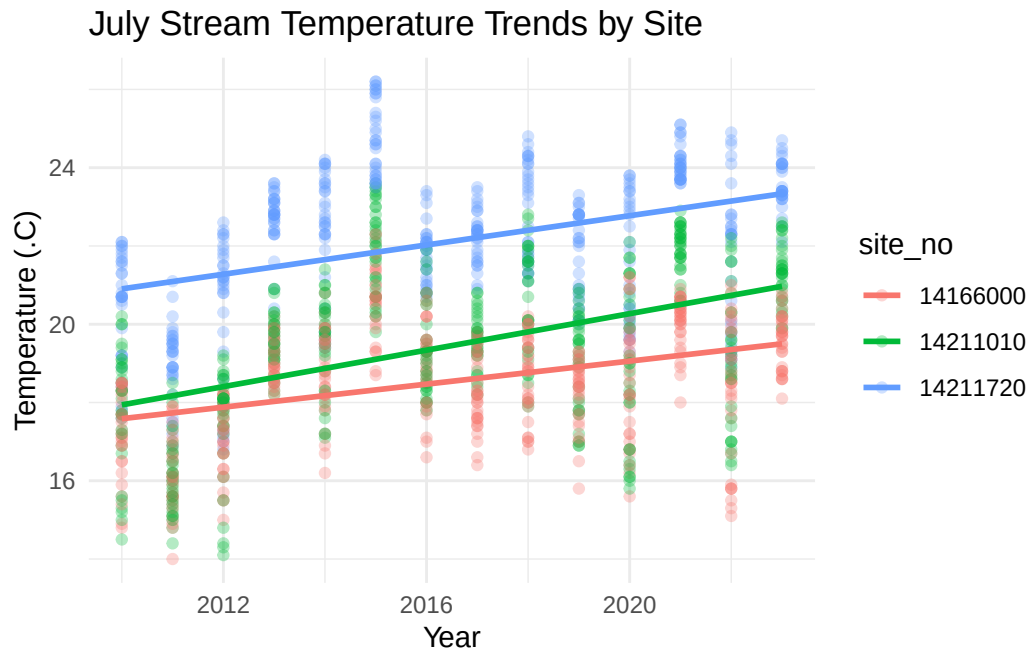
Overall, warming trends were stronger in July than in August, suggesting that peak summer temperatures may be increasing more rapidly.

July Temperature Trends

The figure below shows July stream temperature trends for each monitoring site from 2010-2023.

```
july_data <- temp_data %>%
  filter(month == "Jul")

ggplot(july_data, aes(x = year, y = temp_C,
                      color = site_no)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "July Stream Temperature Trends by Site",
    x = "Year",
    y = "Temperature (°C)",
  ) +
  theme_minimal()
```



July Interpretation

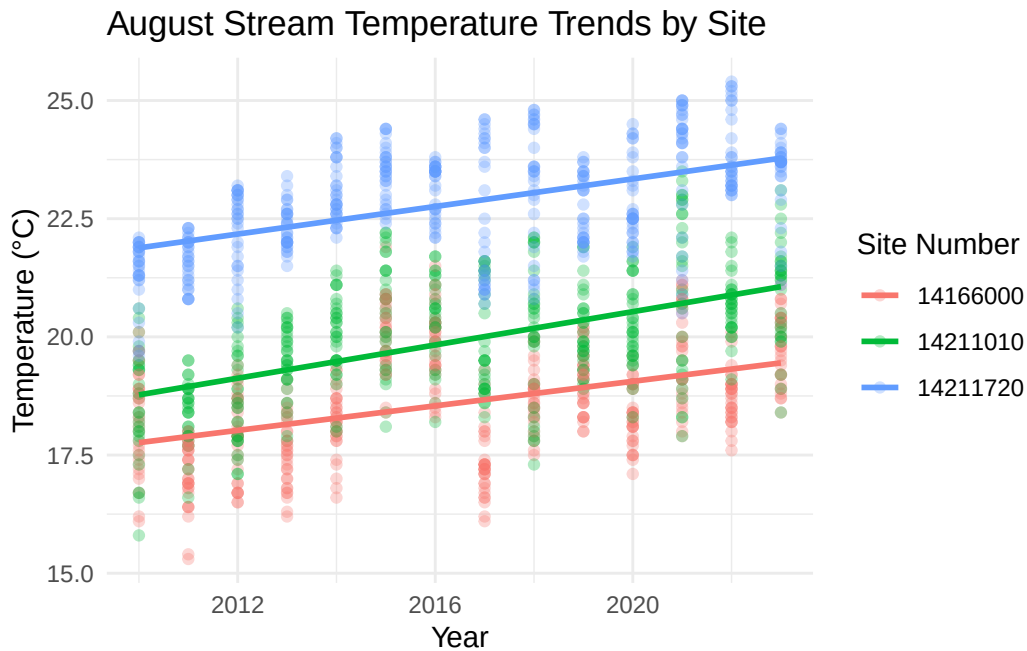
July temperatures showed a strong positive warming trend over time at all monitoring sites. Site 14211010 exhibited the most rapid warming trend, while site 14211720 showed a smaller and non-significant difference from the baseline site.

August Temperature Trends

The figure below shows August stream temperature trends for each monitoring site.

```
aug_data <- temp_data %>%
  filter(month == "Aug")

ggplot(aug_data, aes(x = year, y = temp_C, color = site_no)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "August Stream Temperature Trends by Site",
    x = "Year",
    y = "Temperature (°C)",
    color = "Site Number"
  ) +
  theme_minimal()
```



August Interpretation

August temperatures also increased significantly over time, although warming rates were slightly lower than those observed in July. Site 14211010 again showed the strongest warming trend among the monitoring sites.

Comparison of July and August Trends

The figure below compares average July and August temperatures across all monitoring sites over time.

```
summary_data <- temp_data %>%
  filter(month %in% c("Jul", "Aug")) %>%
  group_by(year, month, site_no) %>%
  summarise(
    mean_temp = mean(temp_C, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(summary_data,
  aes(x = year,
    y = mean_temp,
    color = site_no,
```

```

    linetype = month)) +

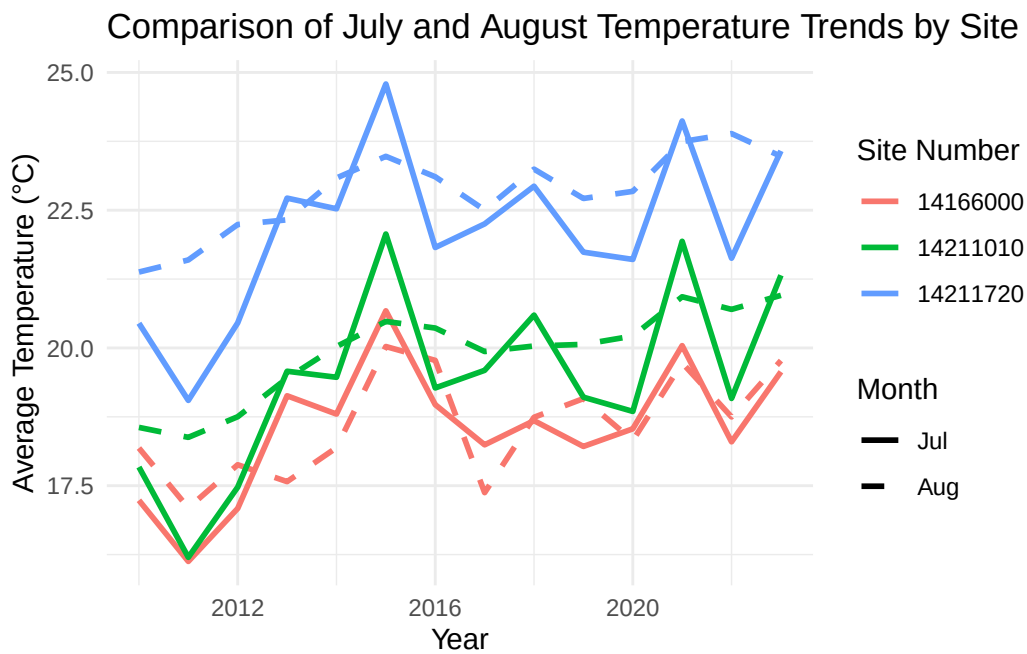
geom_line(size = 1) +

labs(
  title = "Comparison of July and August Temperature Trends by Site",
  x = "Year",
  y = "Average Temperature (°C)",
  color = "Site Number",
  linetype = "Month"
) +

scale_linetype_manual(
  values = c(
    "Jul" = "solid",
    "Aug" = "dashed"
  )
) +

theme_minimal()

```



Comparison Interpretation Both July and August temperatures increased over time across

all three monitoring sites. July temperatures were consistently higher than August temperatures, and site 14211010 showed the strongest warming trend over the study period.

Conclusion

The analysis indicates the stream temperatures in the Willamette River have increased significantly between 2010 and 2023, particularly during the summer months.

July exhibited the strongest warming trends, and one monitoring site (14211010) showed significantly faster warming than the others. Increasing stream temperatures may have important ecological implications, including increased thermal stress for aquatic organisms and reduced water quality during summer months.

These results suggests that continued monitoring of stream temperatures is important for understanding climate-related impacts on river ecosystems.